Gabriel Ramirez *April 9, 1983*

luga.ramirez@gmail.com • lugaramirez (*LinkedIn*) • lugaramirez (*GitHub*) • Marius Bauerstraat 30-B4 • Amsterdam, 1062AR • The Netherlands

---

Summary

I specialize in simplifying architectures that have grown too complex for the teams that depend on them — diagnosing where the weight is, making the structural call, and seeing it through. Over 20 years I've done this across healthcare, e-commerce, fintech, banking, and public infrastructure, in Paraguay, the US, and the Netherlands. My background starts in electrical and embedded engineering, which trained a different kind of systems instinct: I think in constraints, failure modes, and physical limits before I think in abstractions. When a system starts collapsing under growth, the root cause is almost always a physical constraint the architecture stopped respecting — spotting that gap, e.g., latency, memory pressure, is what the first half of my career trained me to see.

Experience

bol. — Utrecht, The Netherlands

Software Engineer — *Nov '23 – Present*

The core problems: a tightly-coupled legacy monolith blocking cross-team independence, an inherited culture that treated legacy stability as a reason to avoid tests, and an emerging need to govern AI-assisted development responsibly at scale.

- Migrated login handling out of the monolith; chose an event-based design over direct service coupling, enabling other teams to add post-login behaviour autonomously without coordination overhead.
- Demonstrated test-gated automatic releases on a legacy service; the approach spread organically, leading to an invitation to join the 4-person company-wide quality community.
- Owns quality and security constraints for AI-assisted code generation (Claude) company-wide; introduced promptfoo as a pre-release gate for Claude Code skills before company-wide distribution.
- Maintains systems at scale (20M+ users, 95k+ req/s) across Go, Java, Kotlin, and Python — edge router, GraphQL connectors, and the monolith-to-microfrontend migration.

Quin — Amsterdam, The Netherlands

Tech Lead — *Sep '21 – Oct '23*

The core problems: a microservices architecture collapsing under its own weight — a feature as simple as showing a doctor their patient list took 6 to 8 months to ship, infrastructure costs exceeded $2M annually, deployments were quarterly, and onboarding a new developer took nearly a year.

- Diagnosed the shared commons library as the most immediate drag — accumulated RestTemplate clients for every service in the landscape, loaded by every microservice regardless of need; stripped it down, cutting individual service footprint by 20%+.
- Standardized Git workflows, code review practices, and CI/CD pipelines across all backend teams.
- Bet on a modular monolith over the prevailing microservices orthodoxy; consolidated 70+ services into a single event-driven architecture with 5 bounded domains and 3 anti-corruption layers, applying Domain-Driven Design throughout.
- The consolidation cut infrastructure costs from $2M+ to under $300K (85%+ savings), accelerated deployments from quarterly to every merge, and reduced onboarding from nearly a year to days. The company, which had entered bankruptcy, recovered and resumed growth.

Backbase — Amsterdam, The Netherlands

Tech Lead — *May '20 – Aug '21*

The core problems: an integration strategy built around one-off scenario forms with no reusable components across them, and a release process so fragile that every deployment broke something.

- Diagnosed that the scenario forms shared underlying integration components (identity verification, SMS/email, document signing) being reimplemented repeatedly; redesigned the approach around composable connectors — personally built 5, the template grew to 12 with the team, and enabled 20+ connectors after departure.
- Built the Angular frontend components for the connector layer personally — identity verification flows, multi-step onboarding forms, and SMS/email journeys — so each connector was a deployable end-to-end banking journey, not just a backend integration point.
- Diagnosed the release failures as a testing gap; introduced acceptance tests and hands-on testing practices across the team, cutting re-releases from twice weekly to once per quarter across 5 banking clients.

Backend Trainer *Jan '20 – May '20*

Delivered 3 week-long training sessions on Backbase's core banking platform to approximately 30 client engineers before the role pivoted due to the pandemic.

Software Natura — Asuncion, Paraguay

Team Lead *Feb '17 – Dec '19*

The core problems: a Java call center application that couldn't communicate with a COBOL mainframe (one month, declared impossible by everyone else), followed by a request to "update" the call center that warranted a full architectural diagnosis.

- Built a C-based Linux kernel module to bridge the Java/Struts application to the COBOL mainframe — the only viable path across a hard process boundary — and delivered within the one-month deadline.
- When asked to update the call center, diagnosed it as beyond patching; made the call for a full revamp in Spring Boot and Angular with domain-driven boundaries, and brought in a data specialist for Apache Spark search rather than attempting it without domain expertise.
- Diagnosed the Paraguay-side team as lacking process and standards; introduced structured workflows and code quality practices that enabled a 7-person team to collaborate effectively with 5 US-based developers across time zones.

QA Engineer *May '16 – Jan '17*

Diagnosed a 100k+ line legacy CRM monolith — serving American community colleges in a 50-developer team — as having zero test coverage, meaning bugs only surfaced through weeks of end-user feedback; grew coverage from zero to 60%, shifting detection to build time.

Santco SRL — Asuncion, Paraguay

Software Engineer & Team Lead *Nov '14 – Mar '16*

The core problem: clients with zero tolerance for downtime and no existing software for their most critical operations — delivery reliability, not technical capability, was the actual constraint.

- Structured a team of up to 8 developers around Agile iteration cycles, replacing ad-hoc delivery with predictable, testable releases.
- Delivered 4 process-automation applications on CakePHP and Google Cloud for Paraguay's leading cattle auction house, largest logistics company, and top leather manufacturer — the auction house trusted the software enough to increase bidding events from weekly to every two days.
- Each delivery required learning the client's domain from the ground up before writing a line of code — bidding logic, logistics routing, production cycles. That domain fluency is what made the automation trustworthy rather than just functional.

Early Career — Paraguay, 2005–2012

Before moving full-time into software engineering, I spent roughly seven years building physical systems at national scale: the network and data center infrastructure for Paraguay's 911-equivalent emergency services (a $500M+ resilient optical network spanning major metropolitan areas, supporting tens of thousands of street cameras and real-time coordination of police, fire, and ambulance dispatch); a multi-tenant virtual machine provisioning platform built from first principles before AWS existed, consolidating infrastructure for 15 clients into a shared on-demand model; and network deployments across 500+ banking sites. Running alongside that: the CompuBus research project at Catholic University — embedded C and C++ on microcontrollers with Linux kernel-level communication modules, delivering a working prototype deployed with a real bus company in Asuncion. Thinking in hardware constraints before software abstractions is not background flavour — it is the lens.

Education

University of Leuven — Leuven, Belgium

Masters degree in Nanoscience and Nanotechnology *2012 – 2014*

Area of interest: Bionanotechnology.

Catholic University of Asuncion — Asuncion, Paraguay

Electrical Engineering (discontinued) *2007 – 2010*

Bachelors in Electrical Engineering *2002 – 2006*

Skills

Languages: Java, Kotlin, Go, Python, C, C++, JavaScript, PHP, SQL.

Frameworks & Libraries: Spring Boot, Netflix OSS, CakePHP, jQuery, Angular, React.

Infrastructure & DevOps: AWS, Azure, Google Cloud, Terraform, Docker, Kubernetes, Jenkins, SonarQube, GitLab CI/CD, GitHub Actions, NGINX, Apache, Tomcat, PostgresSQL, MySQL, SQLServer.

Methodologies: Agile, Scrum, Domain-Driven Design, Clean Architecture, Microservices.

Natural languages: Spanish (*native*), Guarani (*native*), English (*full professional proficiency*), German (*limited working proficiency*), French (*elementary proficiency*).

Certifications: System Thinking — MIT xPRO (*2024*), Nanotechnology for Health — imec (*2013*), Introduction to Artificial Intelligence — Udacity (*2009*).